

DuckDB

Description

- In-memory or file-based database that can operate with Python to handle SQL-related workflows very efficiently.
- Provides interoperability with various data processing libraries in Python including `pandas` .

Installation and Setup

- Install DuckDB via PyPi .

```
pip install duckdb
```

Basic Usage - Setup a Connection

```
from os import getcwd
from duckdb import connect

# Path to where you will save the database
db_path = getcwd() + "/path/to/database.db"

# Setup a connection to the database
cursor = connect(database=db_path)
```

Basic Usage - Read CSV Files into Tables

- ... indicates a continuation from the previous section.

```

...

# URL to some public data
csv_url = "https://ashfaq-nsclc-dataset.s3.us-east-1.amazonaws.com/ali_datasets/Electri

# Write your SQL query
query = f'''

    SELECT *
    FROM read_csv_auto("{csv_url}")
    LIMIT 100

...

# Execute SQL using the .sql() method from the cursor
result = cursor.sql(query=query)

print(result)

```

Basic Usage - Convert Table Results into DataFrames

```

...

# Convert the result into a DataFrame
df = result.df()

# Get information about the pandas DataFrame
df.info()

```

Basic Usage - Persist the Table

- table_name could be:
 - An actual table that exists inside DuckDB
 - Pandas DataFrame
 - function that auto infers data from another format
 - .csv : read_csv_auto()
 - .parquet : read_parquet_auto()

```
...
```

```
cursor.sql(  
    query = """  
  
        CREATE OR REPLACE TABLE table_name AS (  
  
            SELECT *  
            FROM table_name  
  
        )  
  
    """  
)
```

Basic Usage - Leveraging Pandas to Persist the Data

```
...
```

```
# Use the .to_sql() method from pandas to save tables into the DuckDb database  
df.to_sql(name='table_name',  
          con=cursor,  
          index=False,  
          if_exists='replace')
```

Basic Usage - Filter the Data

```
...
# Assumption - The table is already in DuckDB
query = """
    SELECT *
    FROM table
    WHERE ...
    LIMIT ...
"""

result = cursor.sql(query=query)

print(result)

# DataFrame
df = result.df()

df.info()
```